

Active Components used in Power Electronics

- Main Transistor Types used in Automotive Applications
 - Bipolar Junction Transistors (BJTs)
 - Metal Oxide Silicon Field Effect Transistors (MOSFETs)
 - Insulated Gate Bipolar Transistors (IGBTs)
- Wide Bandgap Transistors
- Application Examples

CLASSIFICATION OF SEMICONDUCTOR COMPONENTS

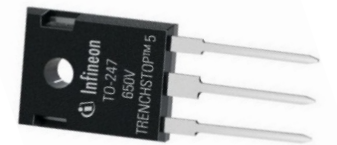
1. Uncontrollable component:

- Has only two terminals and can not be controlled by a control signal
- The on and off states of the device are determined by the circuit
- Diodes



2. Fully-controllable component:

- The on and off states of the device are controlled by control signals.
- Transistors



PACKAGE TYPES

- used in Low&Medium Power Applications
- switching speed of transistors is much higher than the thyristors
- are extensively used in dc-dc and dc-ac converters
- can be turned “ON” and “OFF” by control signals

Surface Mount Device
(SMD)



Through Hole Technology (THT)
e.g. TO 247

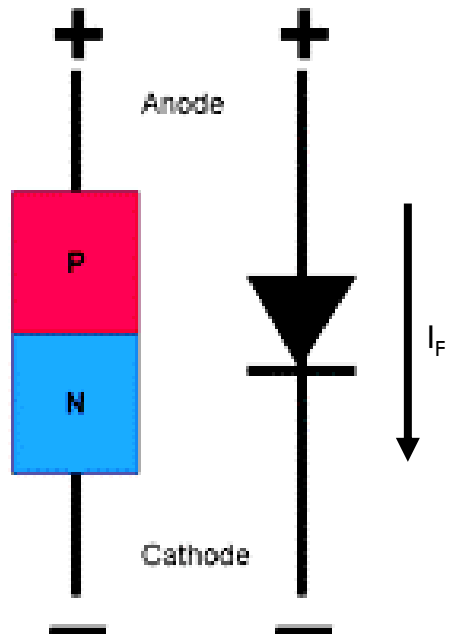


Module Package (high power applications)

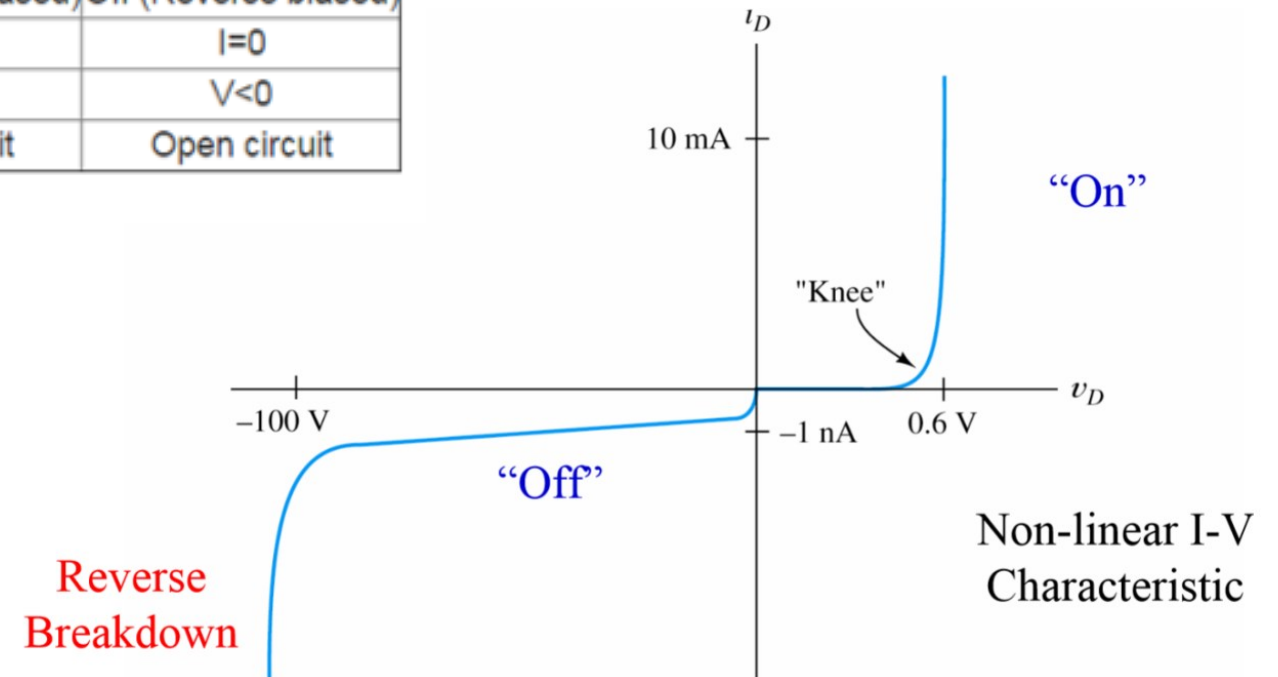


DIODE

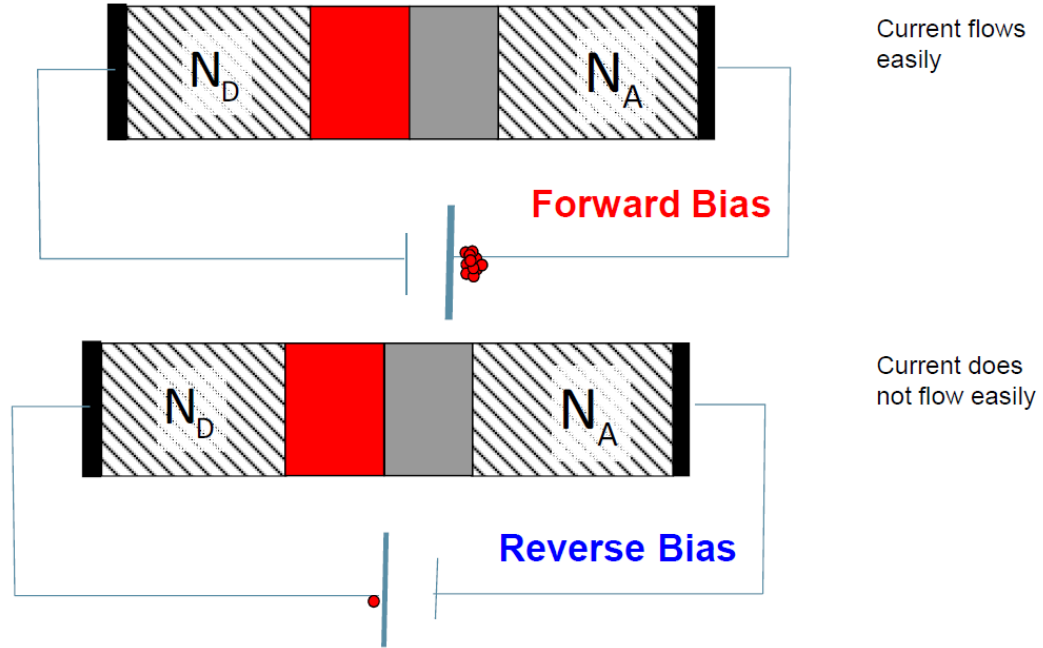
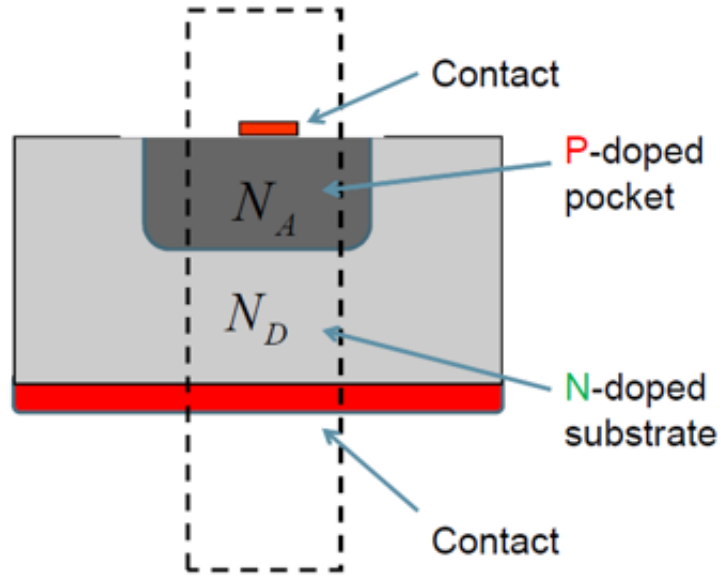
Diodes are made of silicon p-n junction with two terminals, anode and cathode



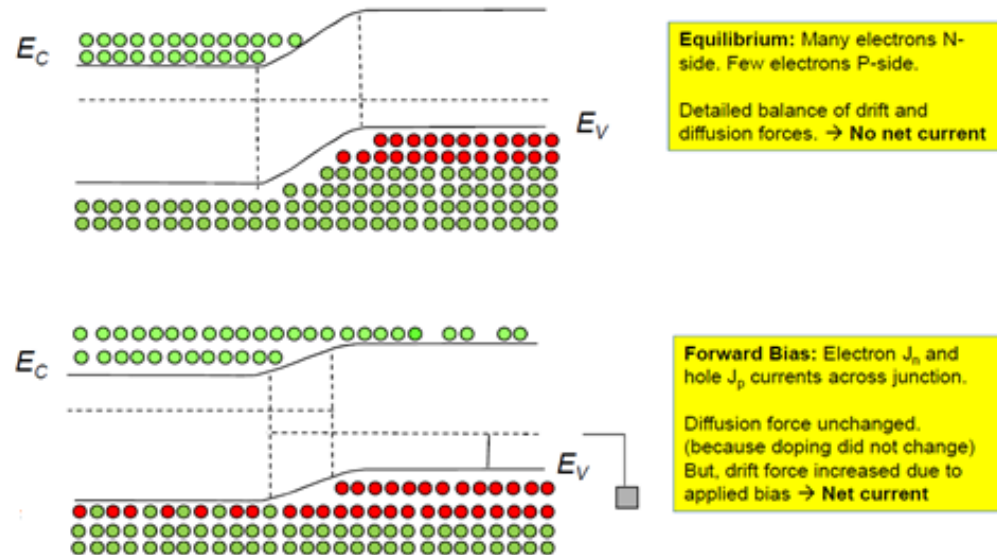
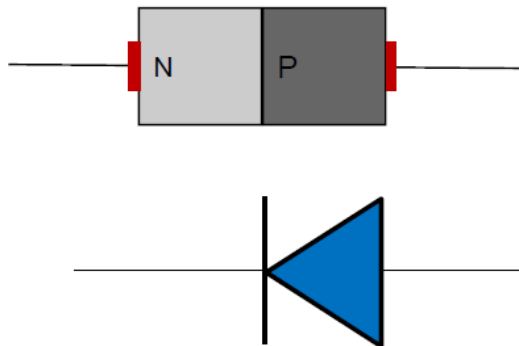
Ideal Diode Characteristics		
Operation Mode	On (Forward biased)	Off (Reverse biased)
Current Through	$I > 0$	$I = 0$
Voltage Across	$V > 0$	$V < 0$
Diode looks like	Short circuit	Open circuit



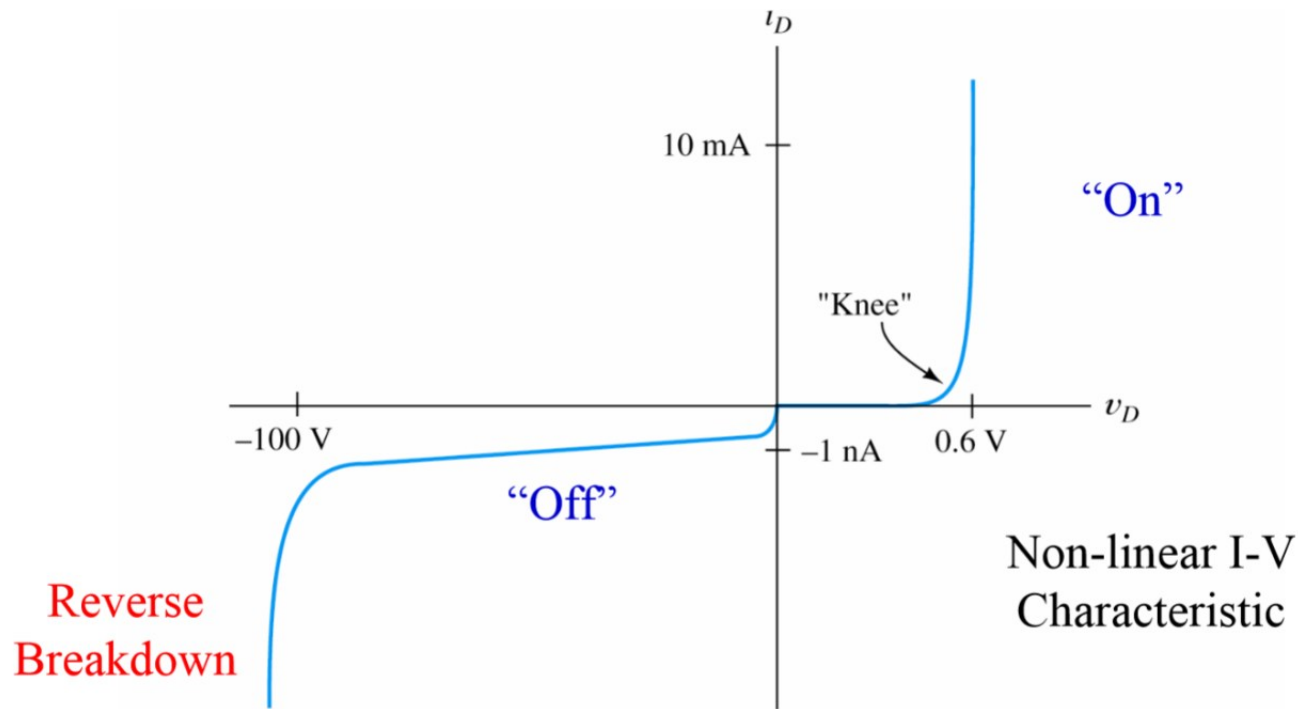
DIODE



Symbol



DIODE



n = Emission coefficient ($1 < n < 2$)
 $k = 1.38 \times 10^{-23}$ J/K is Boltzmann's constant and
 $q = 1.60 \times 10^{-19}$ C is the magnitude of the electrical charge of an electron. At a temperature of 300 K, we have

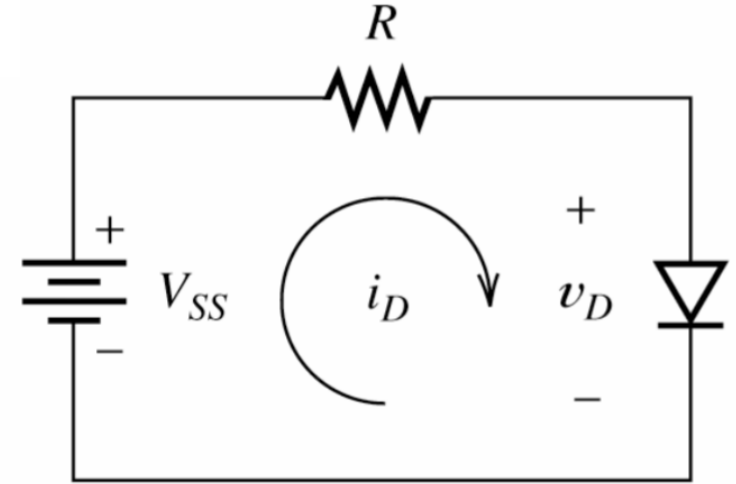
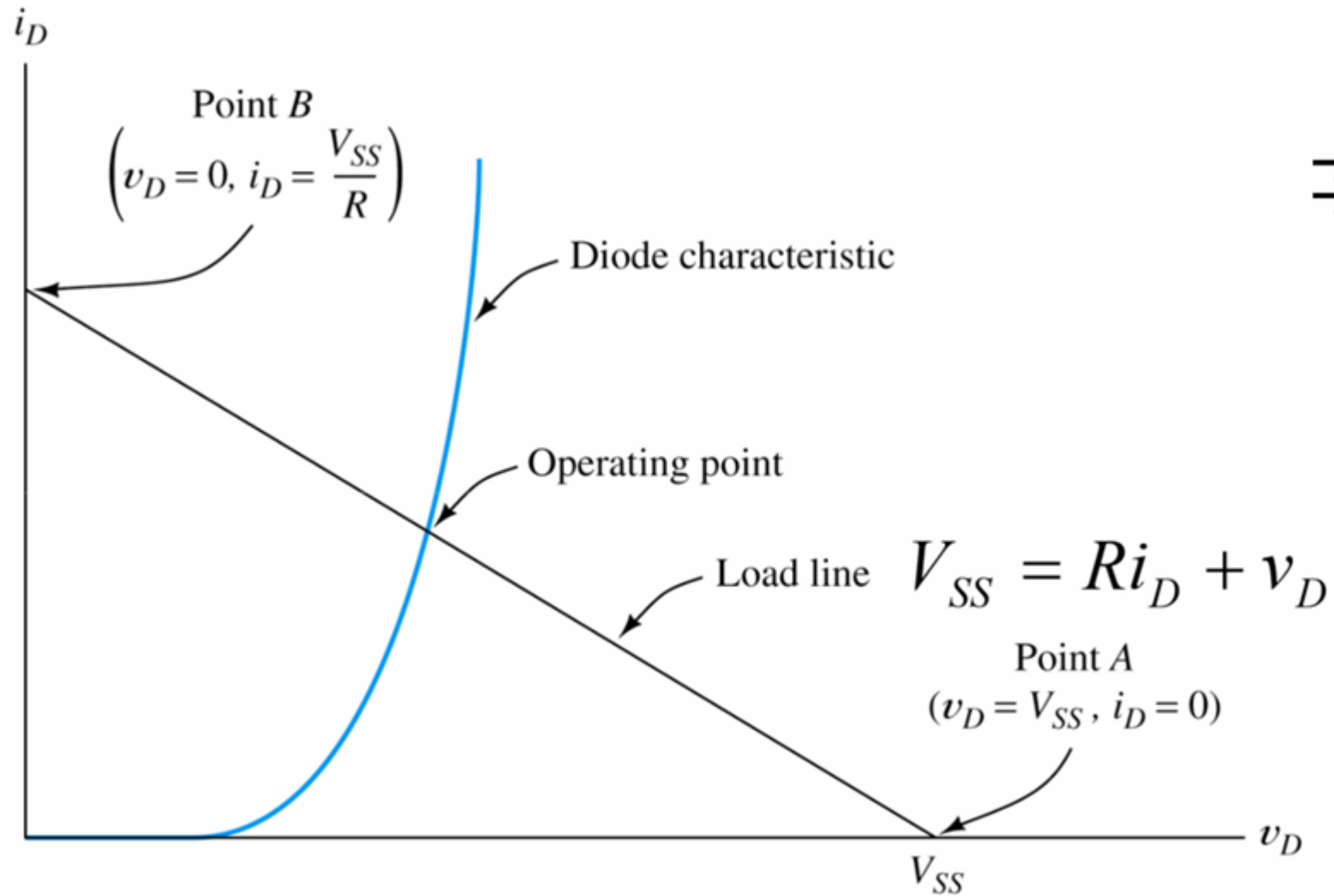
$$V_T \cong 26 \text{ mV}$$

$$i_D = I_s \left[\exp\left(\frac{v_D}{nV_T}\right) - 1 \right] \quad V_T = \frac{kT}{q}$$

$$i_D \approx I_s \exp\left(\frac{v_D}{nV_T}\right) \text{ for } v_D \gg V_T$$

$i_D \approx -I_s$ for $v_D \ll -V_T$ until reaching reverse breakdown

DIODE



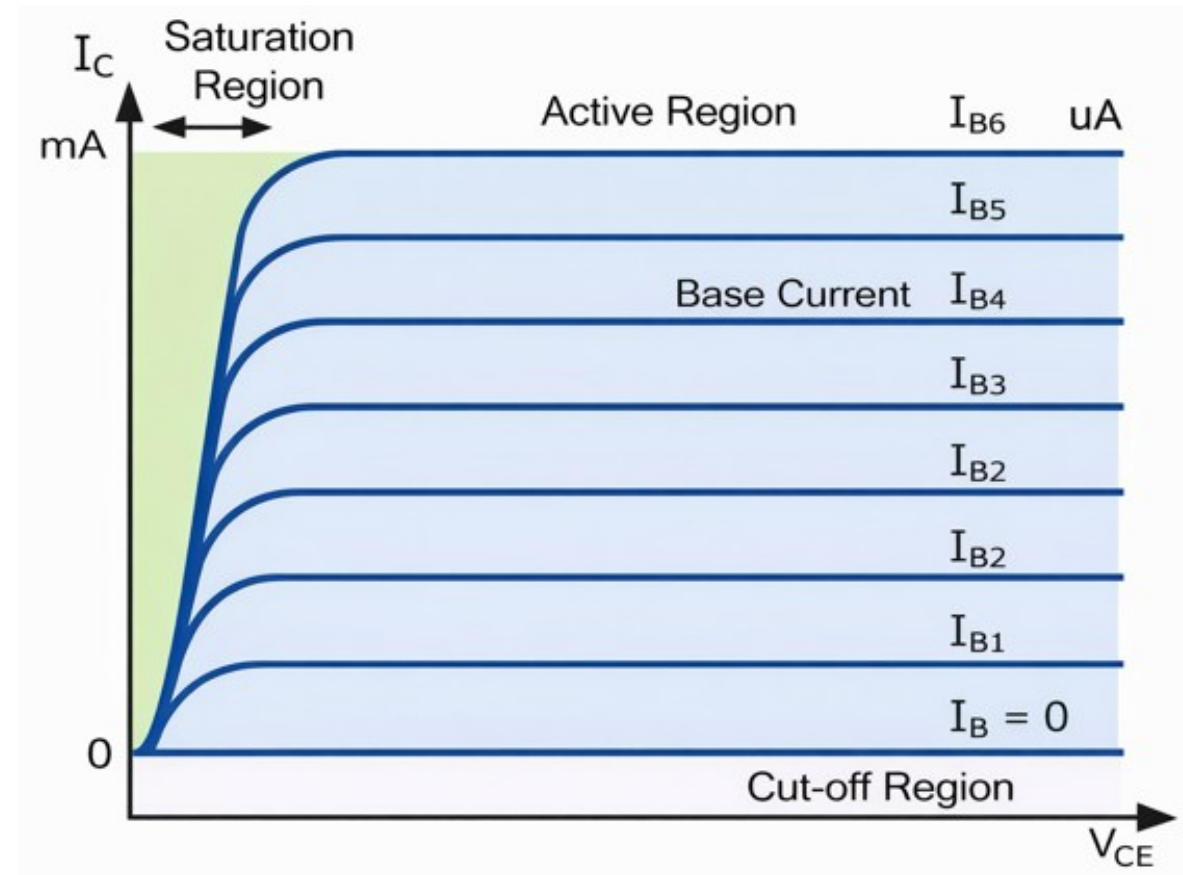
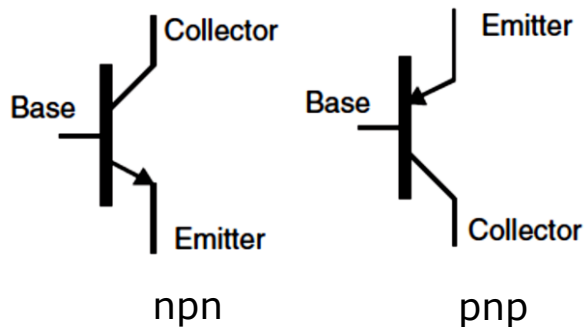
DIODE

- **PN junction diode:** allows current to flow in one direction and is mainly used for rectification in power supplies (already covered in the previous slides)
- **Zener diode:** operates in reverse breakdown and is used for voltage regulation and reference applications
- **Schottky diode:** has a low forward voltage drop and fast switching speed, suitable for high-frequency and power applications
- Light-Emitting Diode (LED): converts electrical energy into light and is used for indication and illumination → can also be used in isolators, to be covered in the next lectures
- Photodiode: generates current when exposed to light and is used in sensing and detection systems → can also be used in isolators, to be covered in the next lectures

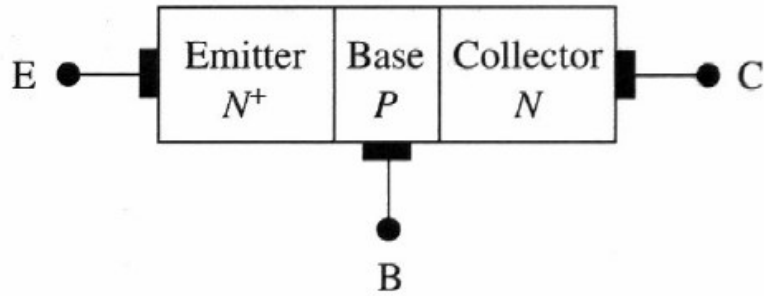
Diode	Zener Diode	Schottky Diode
		

BJT

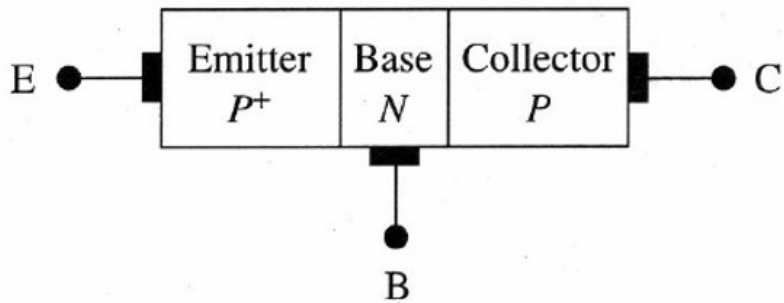
- BJT is a current controlled device
- Has a large blocking voltage in the off state
- Has a high current carrying capability in the on state
- Input impedance is very low => input current is microamps or milliamps in order
- Current-driven => longer switching times
- Expensive and complex base drive circuit => not popular in new products



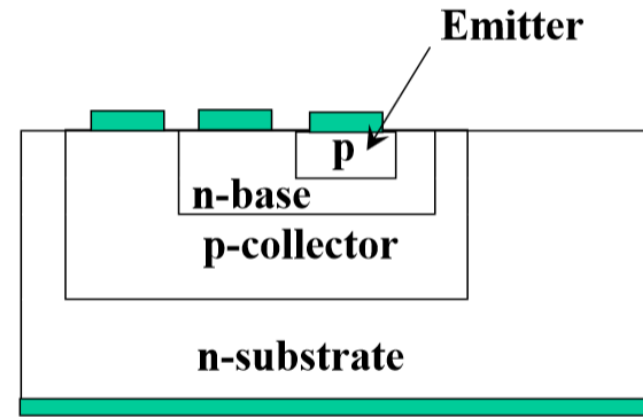
BJT



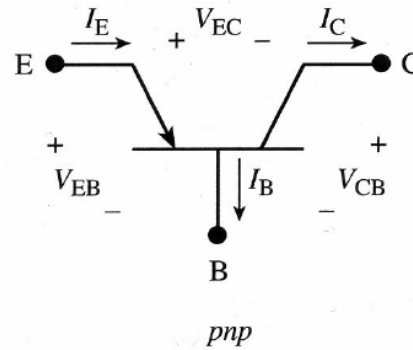
npn



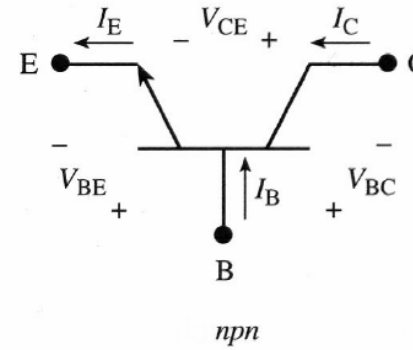
pnp



Cross-section of a pnp transistor



pnp



npn

Current Gain in Forward-Active Region

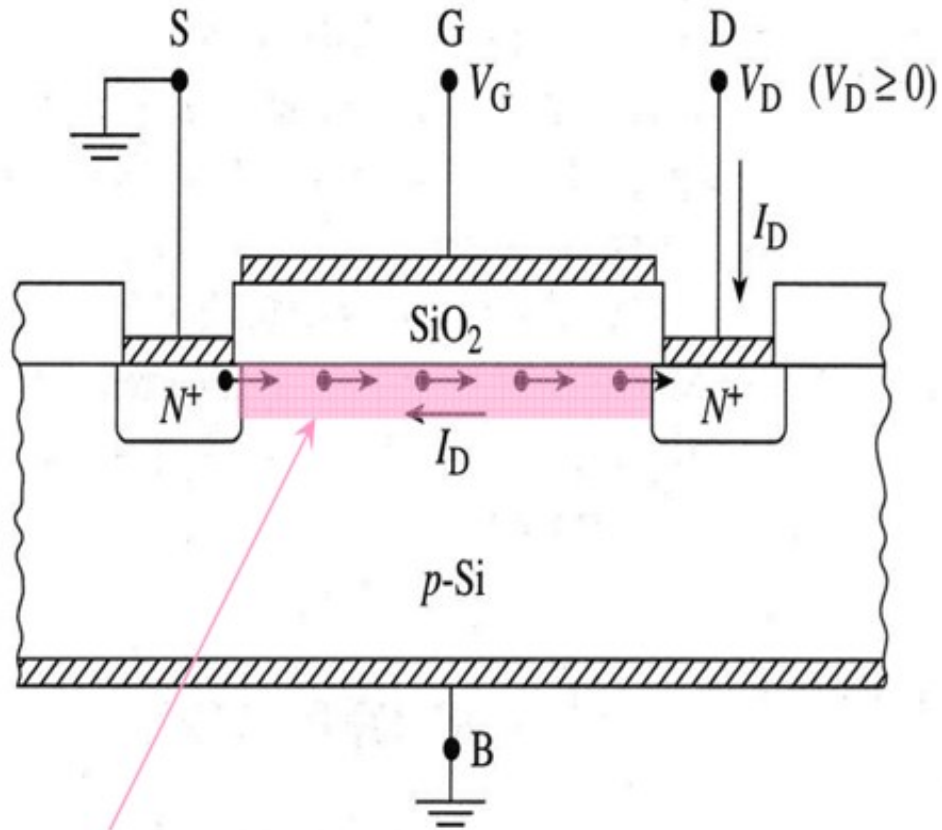
$$\beta = \frac{I_C}{I_B}$$

$$I_E = I_B + I_C$$

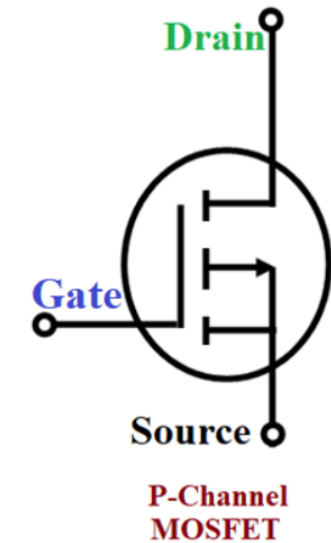
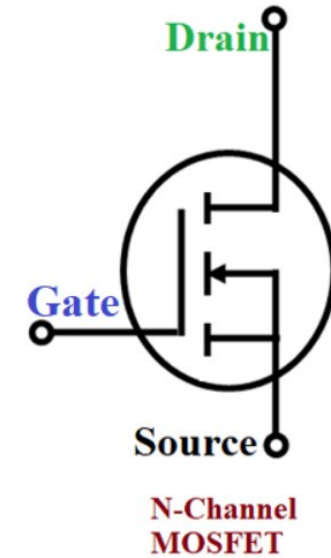
$$V_{EB} + V_{BC} + V_{CE} = 0$$

MOSFET

Flow of current source from Source to Drain is controlled by the Gate voltage

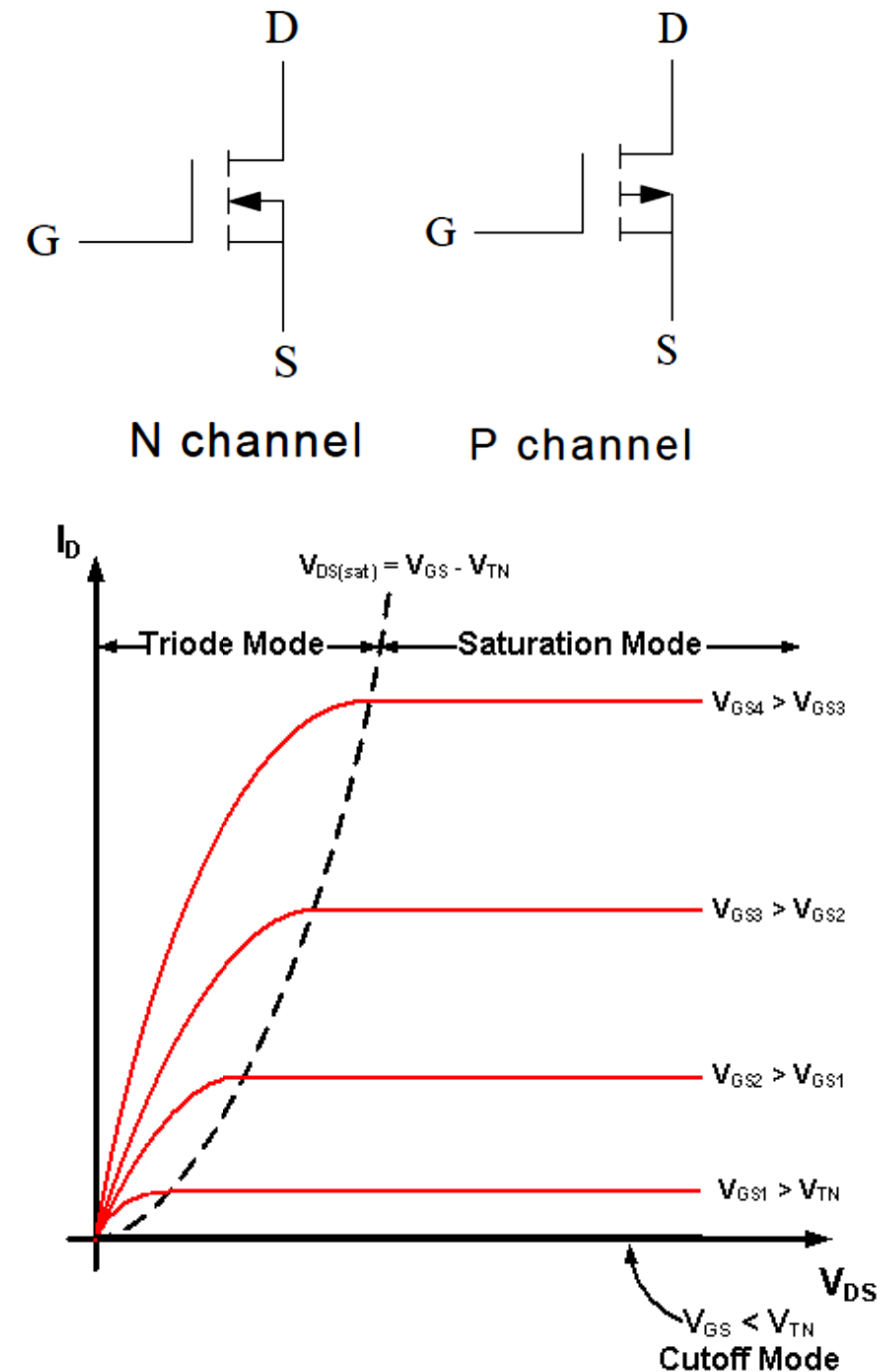


Control by the Gate voltage is achieved by modulating the conductivity of the semiconductor region just below the gate. This region is known as the channel



MOSFET

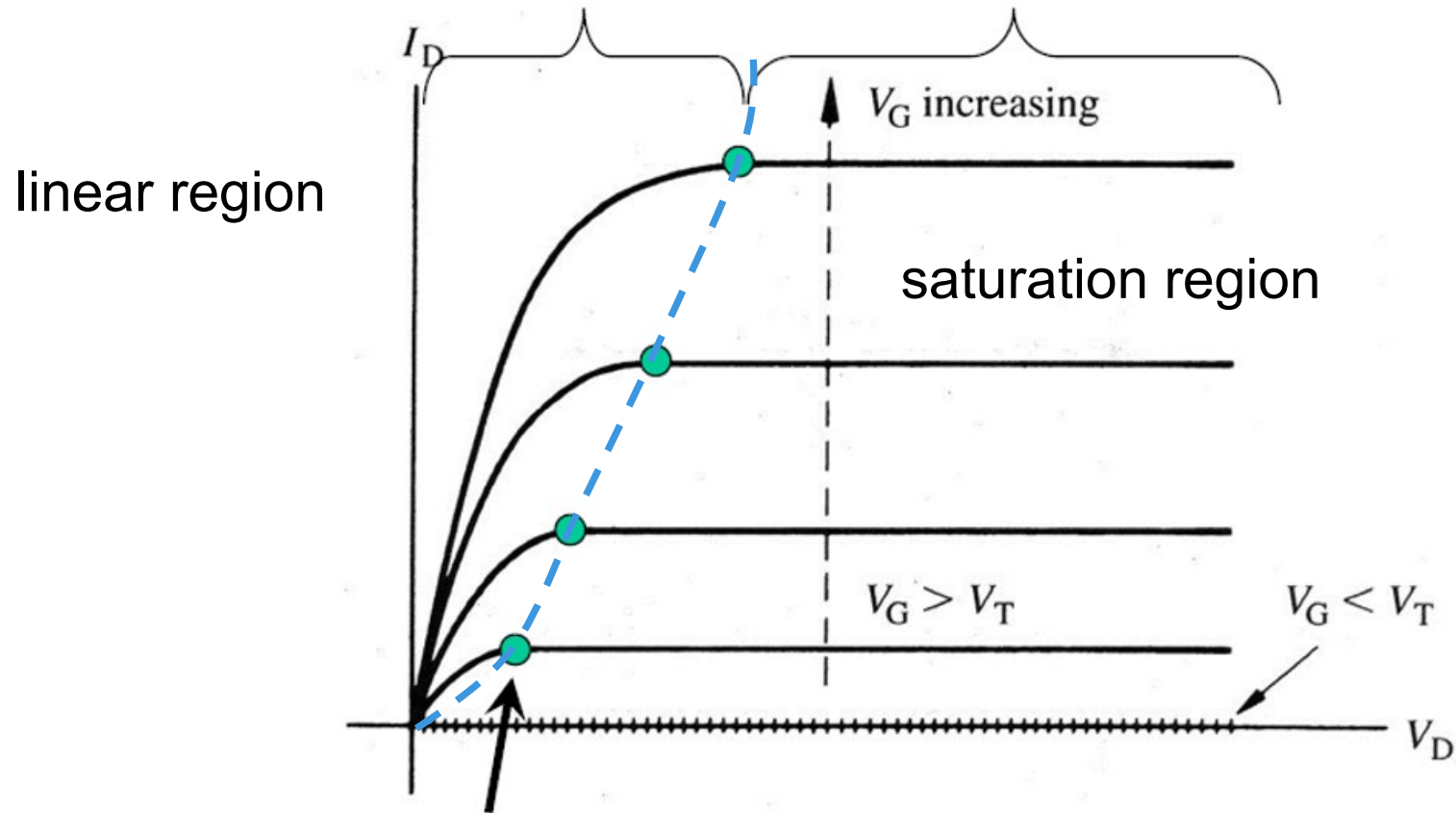
- MOSFET is a voltage controlled device
- Turned 'ON' when a voltage is applied to GS
- Turned 'OFF' by removing the GS voltage
- Voltage driven => faster switching speed => are preferred in applications such as dcdc converters and inverters
- Only voltage drive is required => the drive circuits of MOSFET are very simple
- Have high on-state resistance => for higher currents; losses in the MOSFET's are increased => are used for low/medium power applications
- Input impedance is very high (more than 10^7 Ohms) => input current is approximately zero (picoamps)



MOSFET

$$I_D = \frac{W\mu_n C_{ox}}{L} \left[(V_{GS} - V_T)V_{DS} - \frac{V_{DS}^2}{2} \right]$$
$$0 \leq V_{DS} \leq V_{Dsat} \quad \text{and} \quad V_{GS} \geq V_T$$

$$I_D = I_{Dsat} = \frac{W\mu_n C_{ox}}{2L} (V_{GS} - V_T)^2 \quad V_{Dsat} \leq V_{DS}$$



$$V_{Dsat} = V_{GS} - V_T$$

MOSFET

Linear Region

When the MOSFET is fully enhanced and driven hard on:

$$V_{GS} \gg V_{TH}$$

V_{DS} is forced small by the external circuit

This places the device in the triode (linear / ohmic) region, where:

$$V_{DS} \ll V_{GS} - V_{TH}$$

This is ideal for a switch:

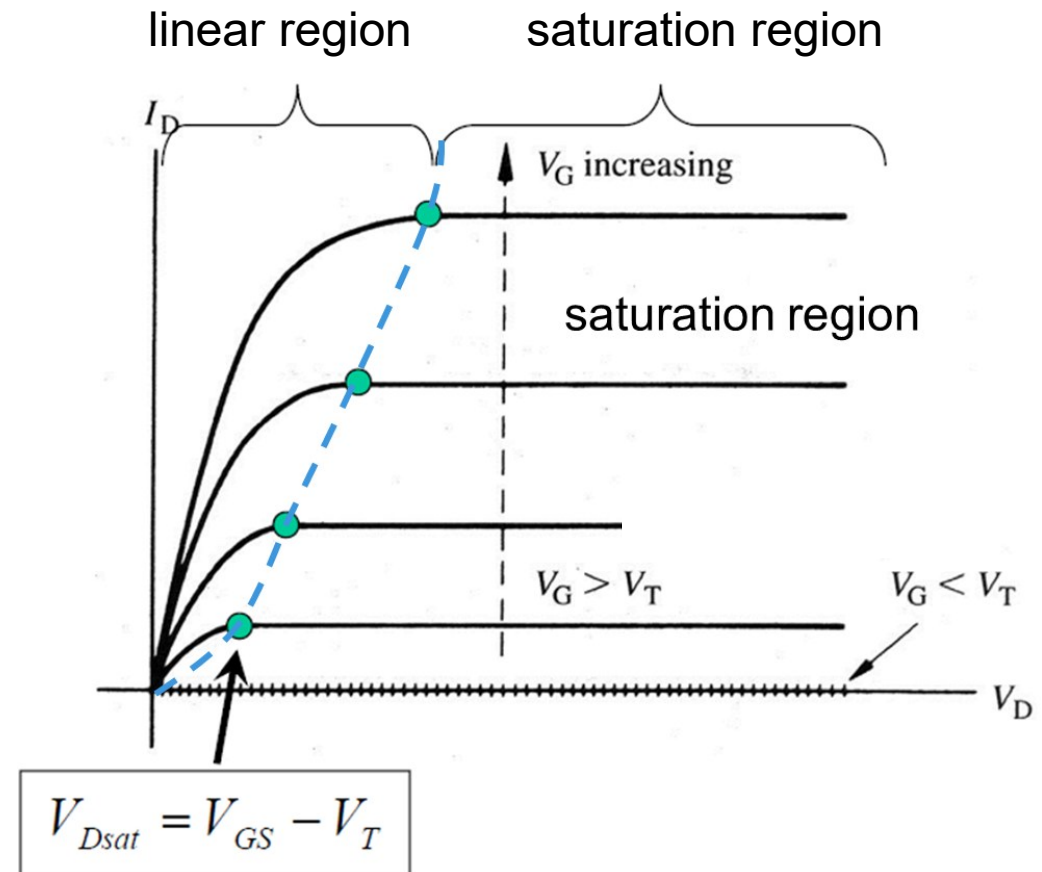
- Low $R_{DS(on)}$
- Low conduction loss $P \approx I_D^2 R_{DS(on)}$
- Minimal heat
- High efficiency

If a switch were operated in saturation:

- V_{DS} would be relatively large
- Power loss $P = V_{DS} \cdot I_D$ would be high
- Efficiency would be poor
- Thermal stress would increase

$$I_D = \frac{W\mu_n C_{ox}}{L} \left[(V_{GS} - V_T) V_{DS} - \frac{V_{DS}^2}{2} \right]$$
$$0 \leq V_{DS} \leq V_{Dsat} \quad \text{and} \quad V_{GS} \geq V_T$$

$$I_D = I_{Dsat} = \frac{W\mu_n C_{ox}}{2L} [(V_{GS} - V_T)^2] \quad V_{Dsat} \leq V_{DS}$$



MOSFET

Saturation Region

$$V_{GS} > V_{TH} \text{ and } V_{DS} \geq V_{GS} - V_{TH}$$

Channel is pinched off near the drain; drain current is mainly controlled by V_{GS} .

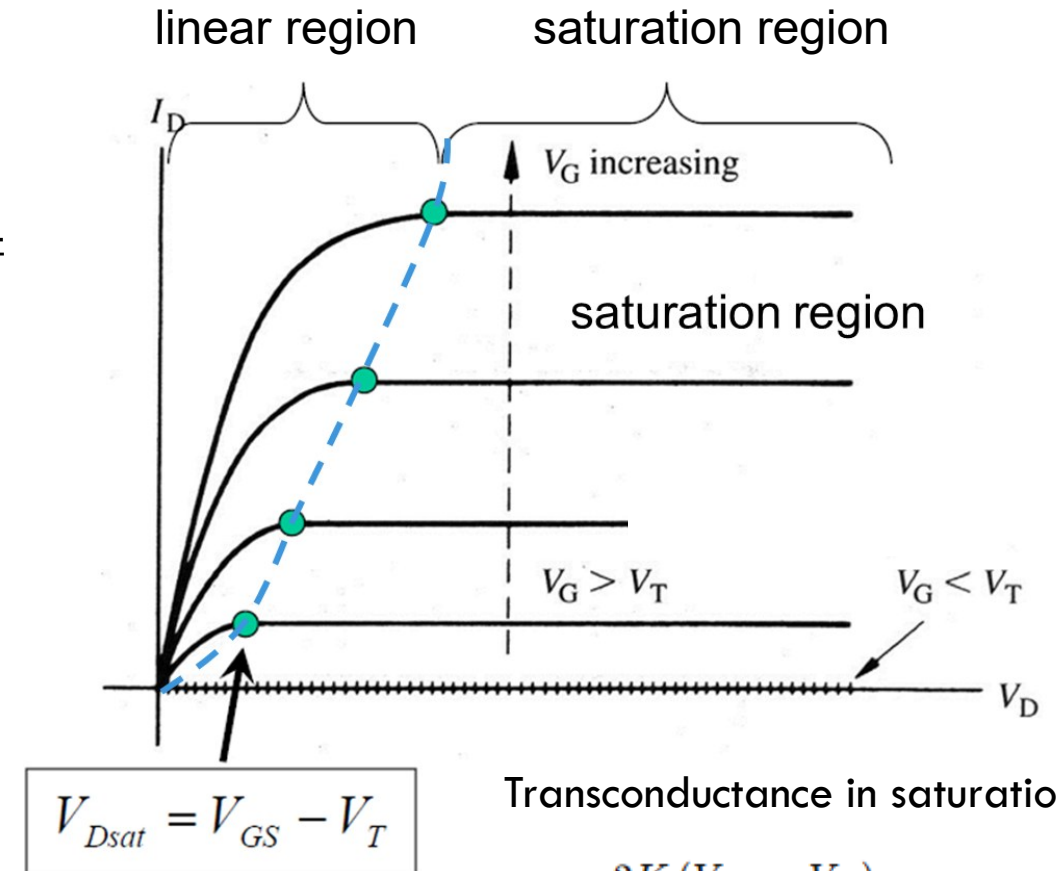
Why amplifiers use saturation region:

In amplification, we want the drain current I_D to be primarily controlled by the gate-source voltage V_{GS} , and to be relatively insensitive to the drain-source voltage V_{DS} .

This gives:

- High transconductance $g_m = \partial I_D / \partial V_{GS}$
- A well-defined, approximately quadratic or linearized relationship between input (V_{GS}) and output current
- The ability to convert small changes in V_{GS} into proportional changes in I_D and hence output voltage via a load

$$I_D = I_{Dsat} = \boxed{K} \left[(V_{GS} - V_T)^2 \right]$$



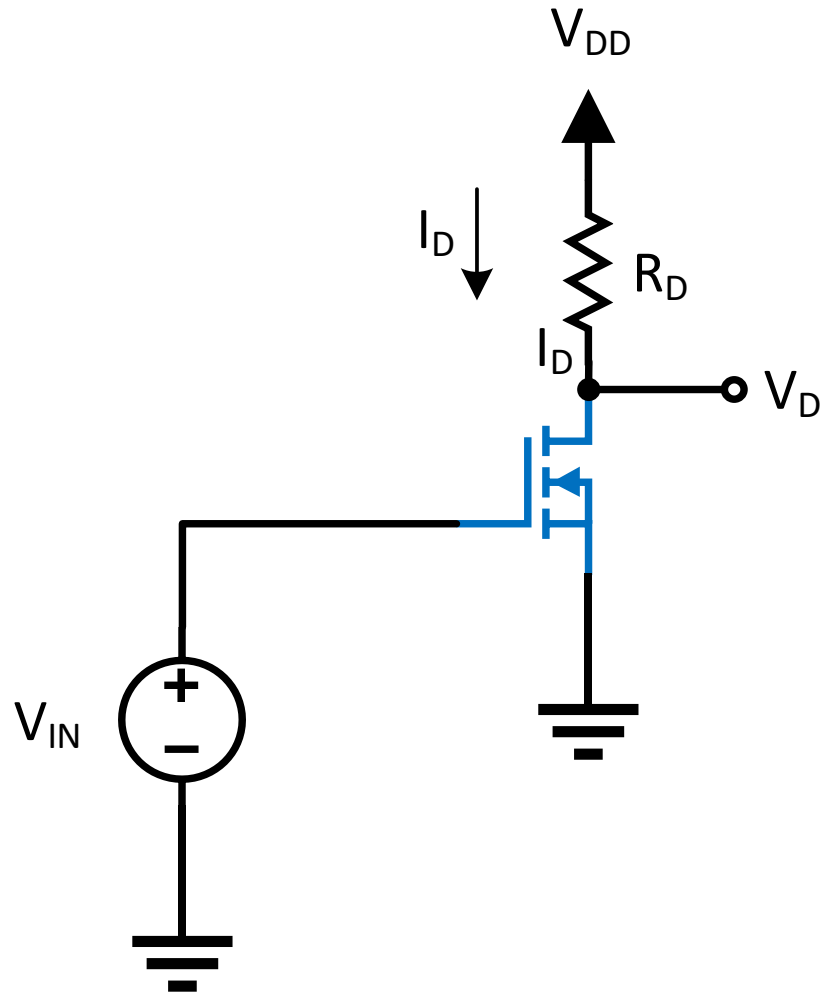
Transconductance in saturation:

$$g_m = 2K (V_{GS} - V_{th})$$

or

$$g_m = 2\sqrt{KI_D}$$

SIMPLE APPLICATION EXAMPLES

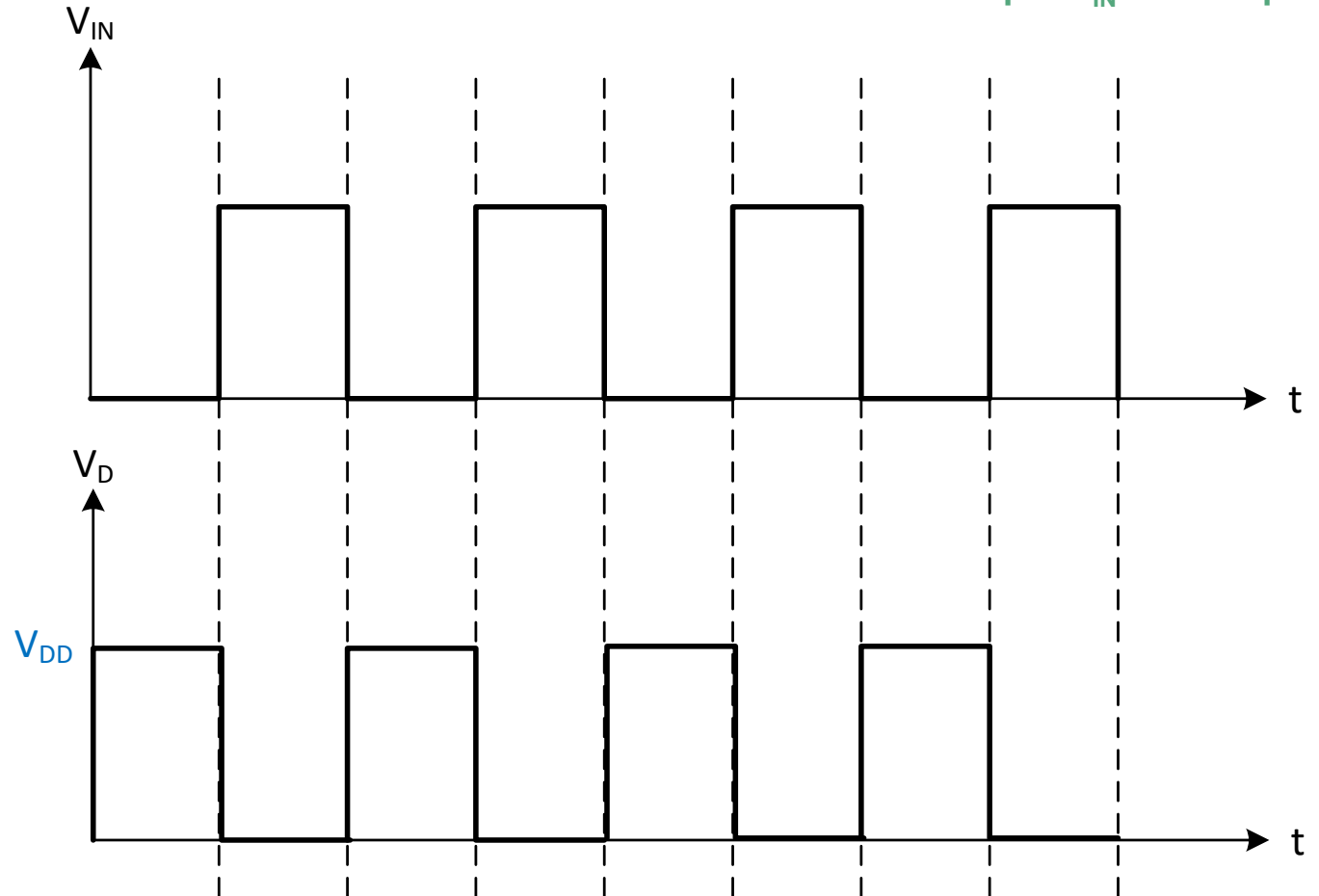


MOSFET acting as a Switch

$$V_D = V_{DD} - R_D I_D$$

$$V_{IN} \gg V_{GS,TH}$$

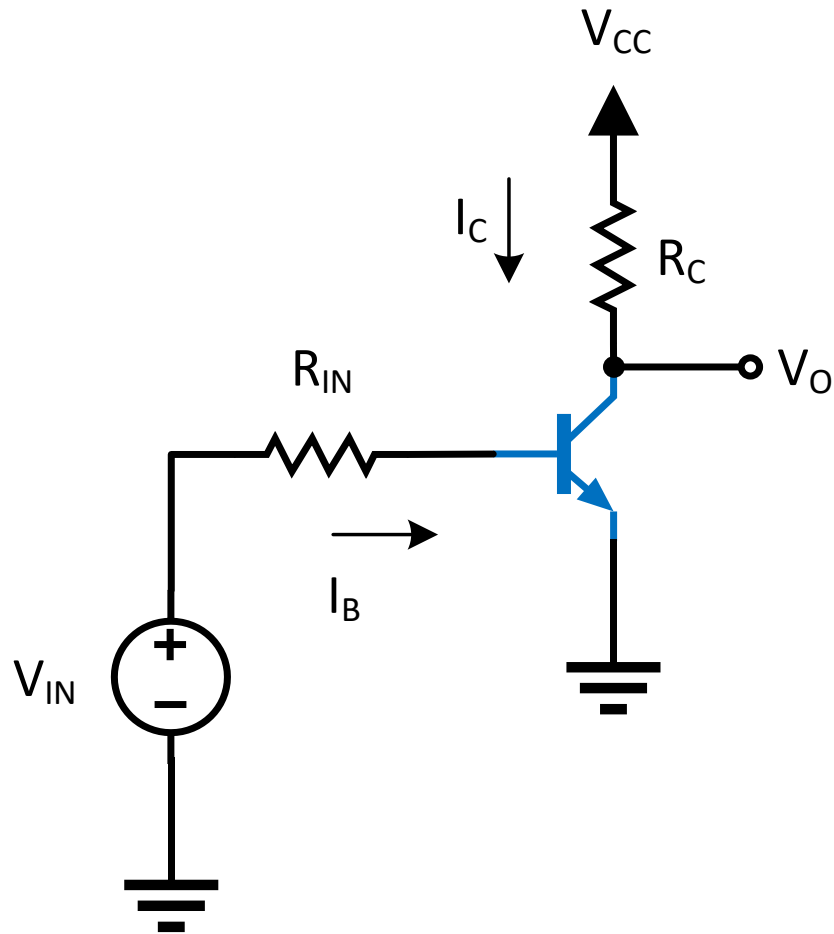
180° Phase Shift between Input V_{IN} and Output V_D



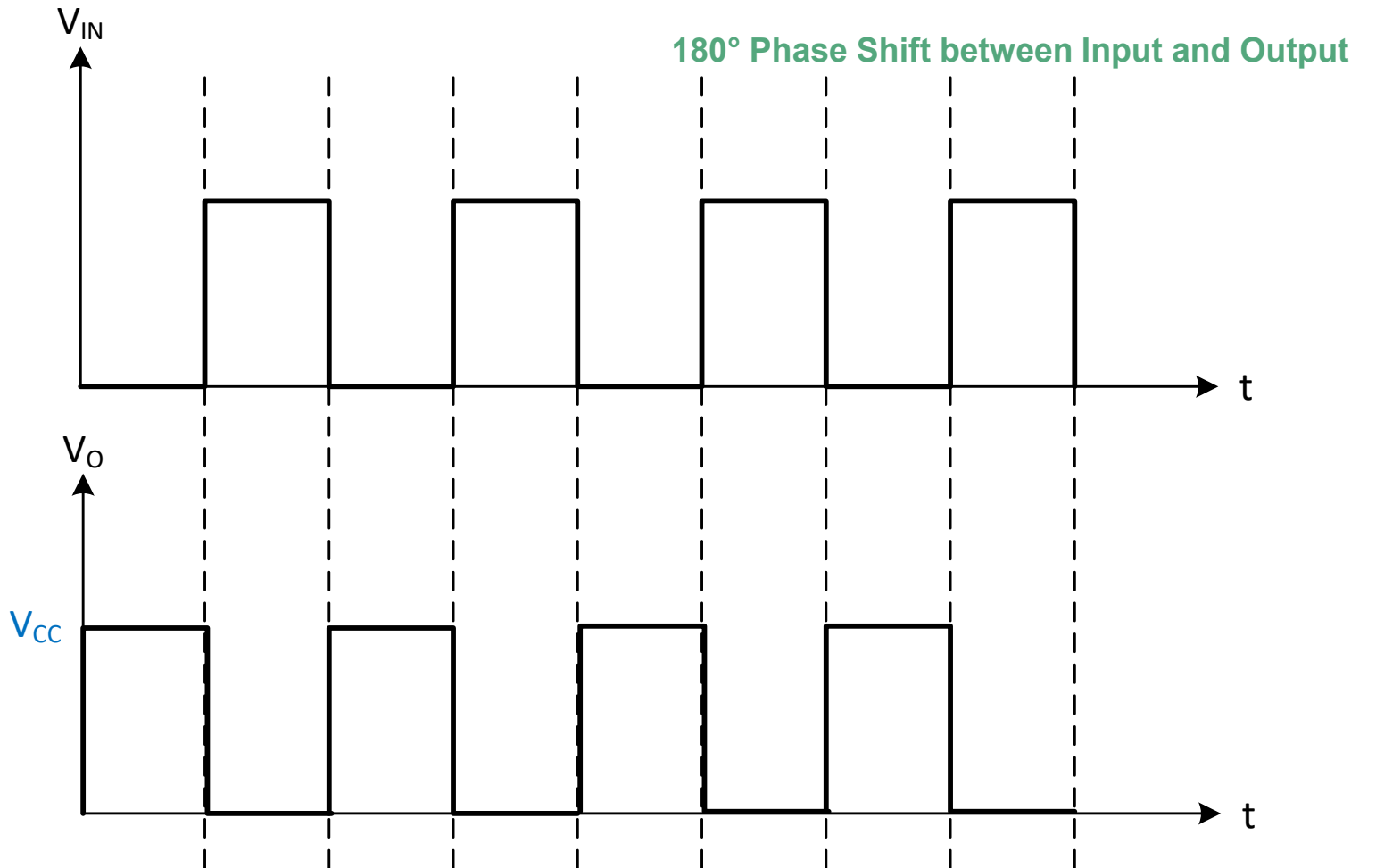
SIMPLE APPLICATION EXAMPLES

$$V_O = V_{CC} - R_C I_C$$

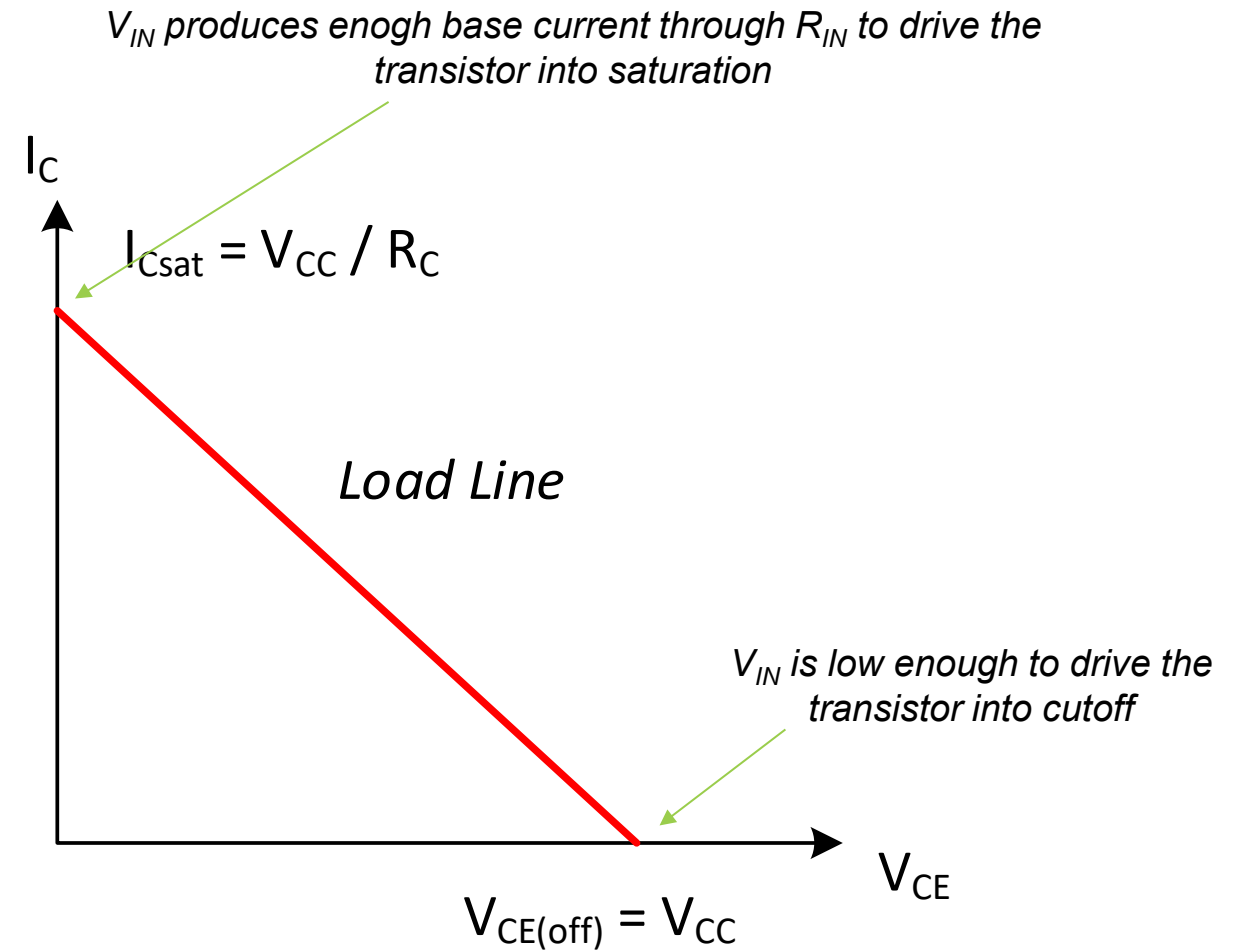
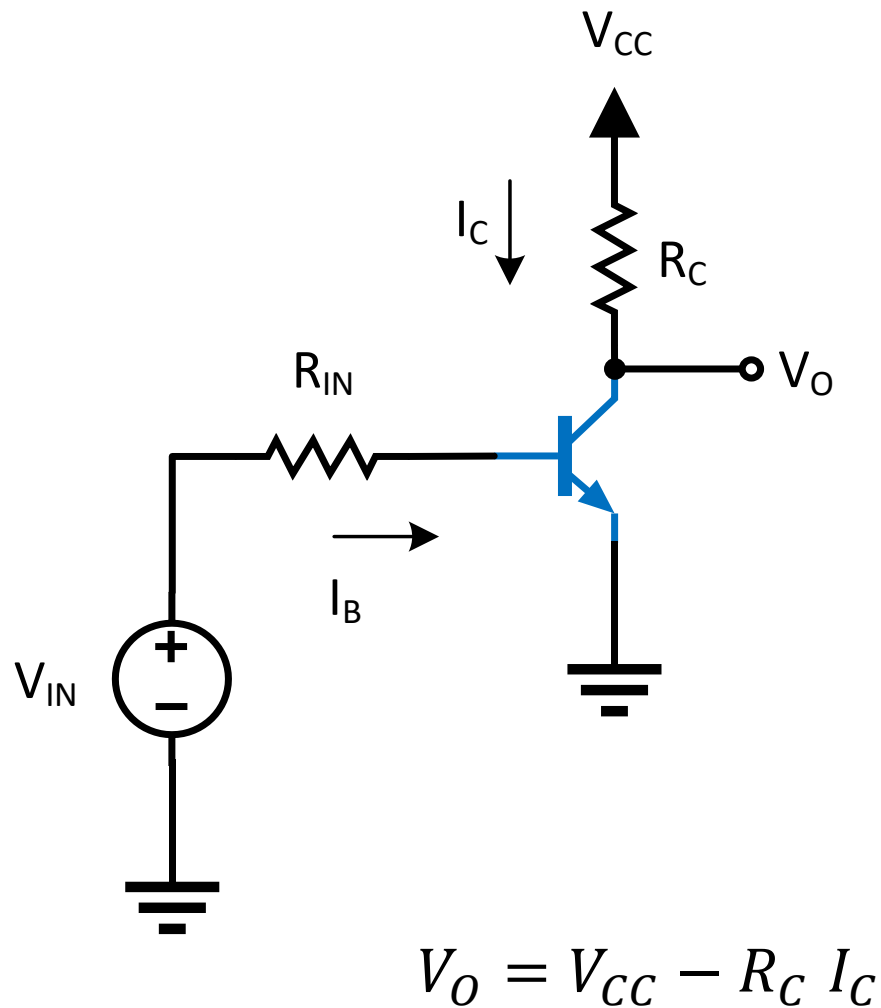
V_{IN} produces sufficient base current through R_{IN} to drive the transistor into saturation



BJT acting as a Switch



SIMPLE APPLICATION EXAMPLES



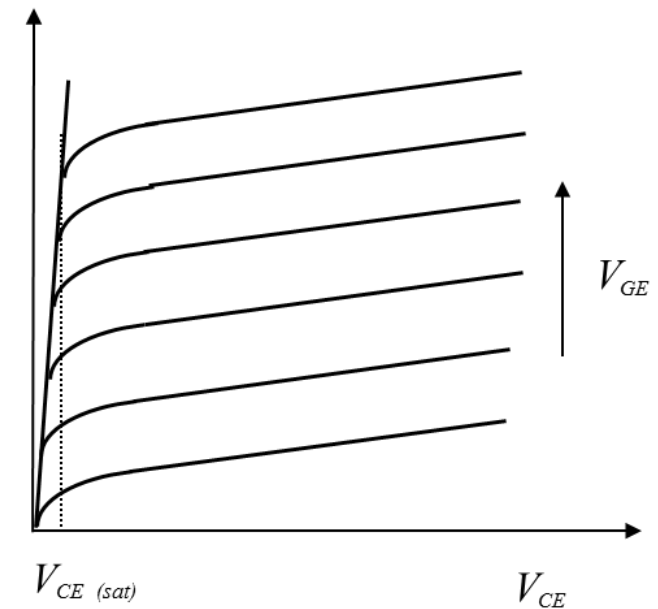
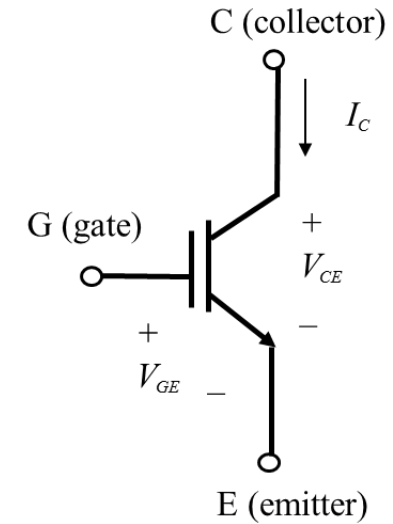
A similar load line can be derived also for MOSFETs

IGBT

- IGBT combines the advantages of BJT's and MOSFET's:
- Gate behaviour similar to MOSFET => easy to turn on and off
- Like BJT, have large current-carrying capability



- Capable of handling large collector-emitter currents with virtually zero gate current drive
- Faster than BJT, but slower than power MOSFET
- Widely used in medium power applications such as DC and AC motor drives, UPS systems, Power supplies for solenoids, relays and contractors

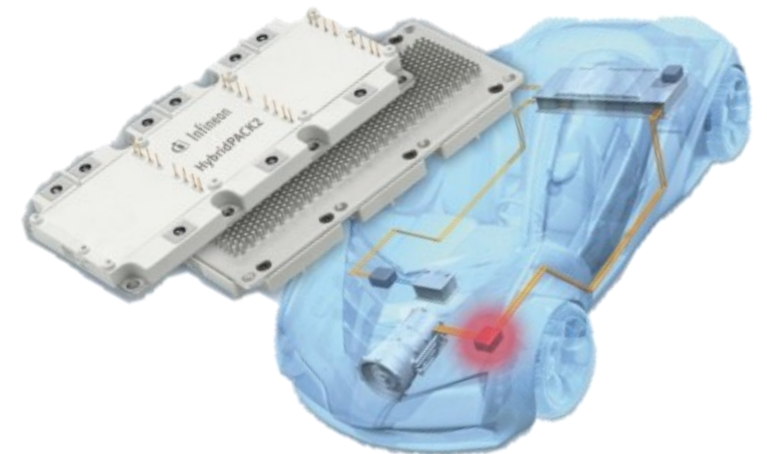


COMPARISON OF BJT AND MOSFET

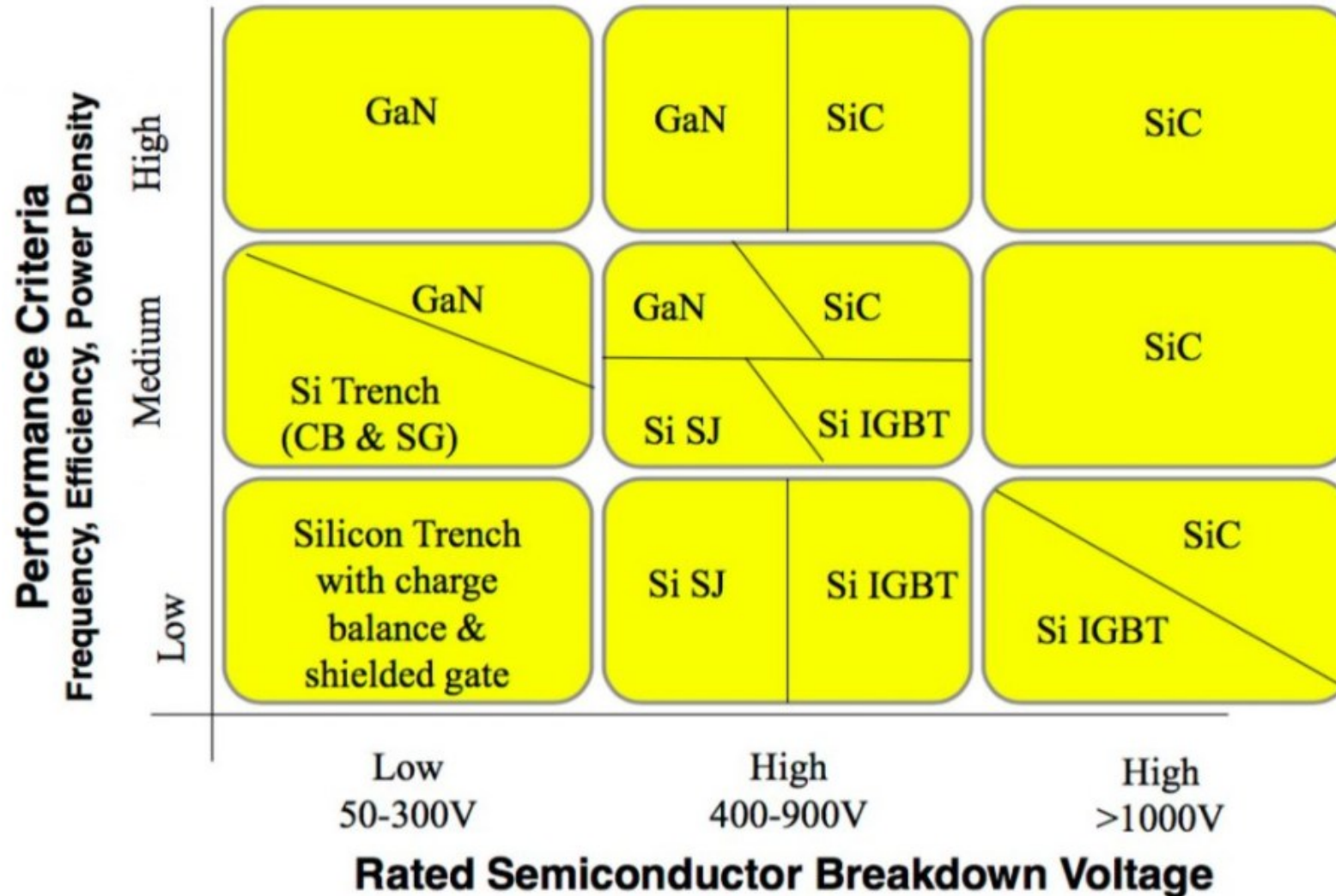
Device Characteristic	Power BJT	Power MOSFET	IGBT
Voltage Rating	High <1kV	High <1kV	Very High >1kV
Current Rating	High <500A	Medium <400A	High >500A
Input Drive	Current	Voltage	Voltage
Input Impedance	Low	High	High
Output Impedance	Low	Medium	Low
Switching Speed	Slow (μ S)	Fast (nS)	Medium
Cost	Low	Medium	High

IGBT

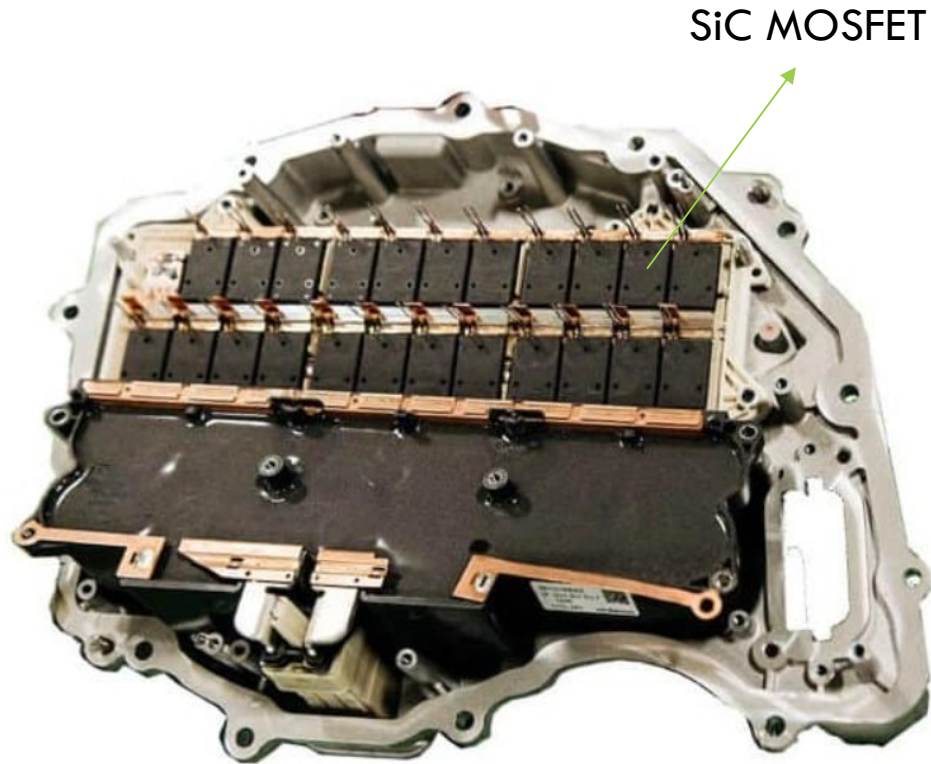
- The main advantages of IGBT over a Power MOSFET and BJT are:
 - It has a very low on-state voltage drop and has superior on-state current density
 - Low driving power and a simple drive circuit due to the input MOS gate structure. It can be easily controlled as compared to current controlled devices such as thyristor, BJT
 - Wide SOA (Safe Operating Area). It has superior current conduction capability.
- The main drawback is:
 - Switching speed is lower than Power MOSFET & higher than BJT.



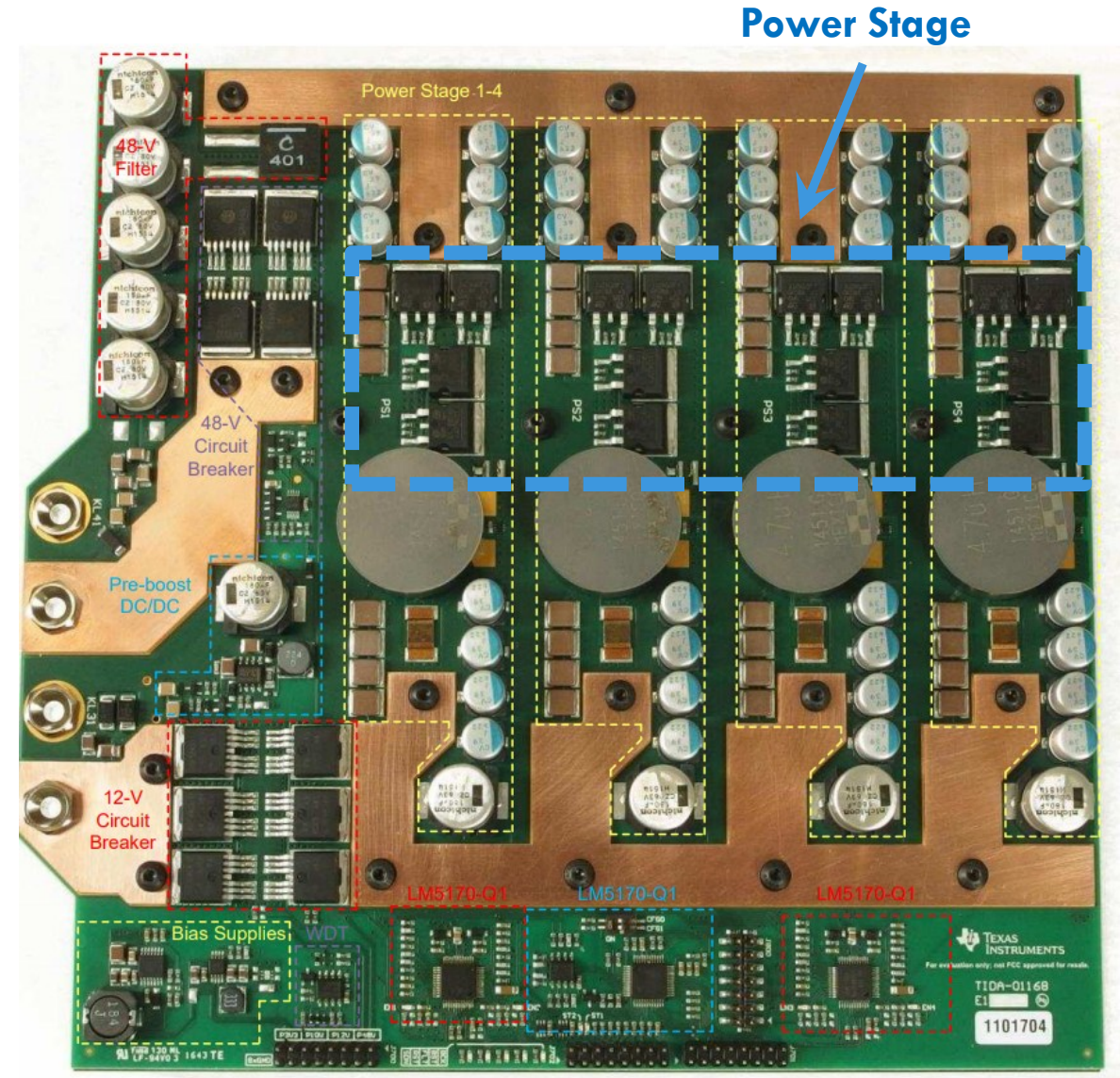
TRANSISTOR COMPARISON



PRODUCT EXAMPLES



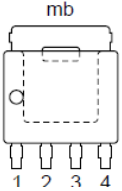
Tesla Model 3 inverter, showing the SiC MOSFET power modules from ST Microelectronics



Bidirectional DC-DC Converter Reference Design for 12-V/48-V Automotive Systems – Texas Instruments

UNDERSTANDING THE DATASHEET OF TRANSISTORS

Table 2. Pinning

Pin	Symbol	Description	Simplified outline	Graphic symbol
1	S	source		
2	S	source		
3	S	source		
4	G	gate		
mb	D	mount base: connected to drain		

MOSFET

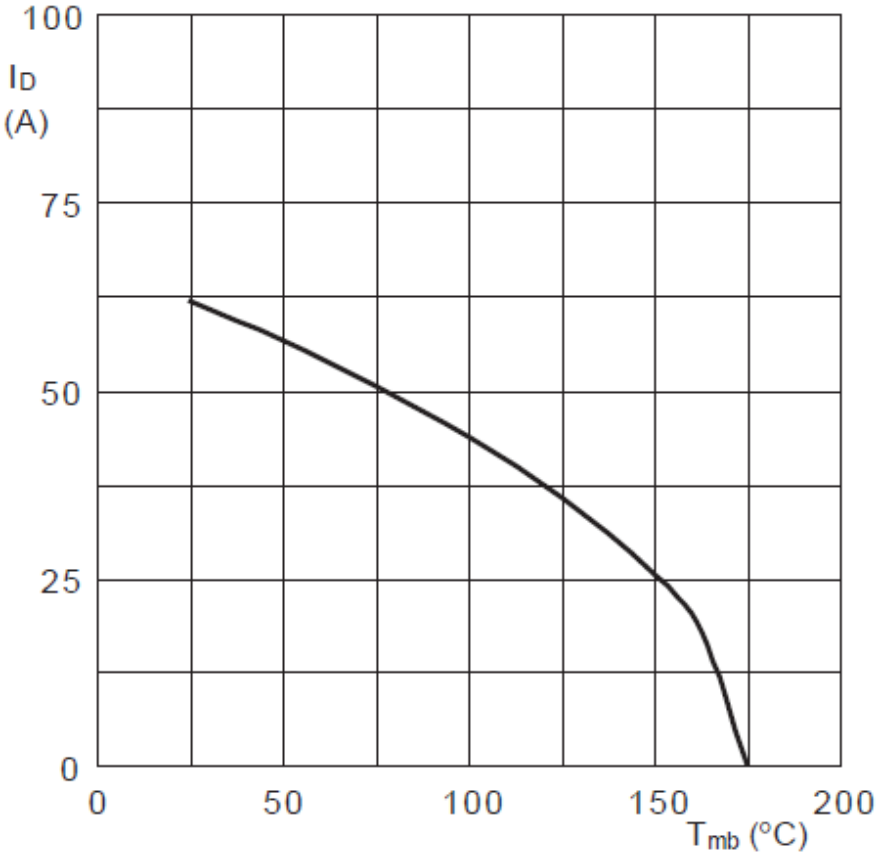
Table 3. Limiting values

In accordance with the absolute maximum rating system IEC 60134

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
V_{DS}	drain-source voltage	$T_j \geq 25\text{ °C}; T_j \leq 175\text{ °C}$	-	-	55	V
V_{DGR}	drain-gate voltage	$R_{GS} = 20\text{ k}\Omega$	-	-	55	V
V_{GS}	gate-source voltage		-20	-	+20	V
I_D	drain current	$V_{GS} = 10\text{ V}; T_{mb} = 25\text{ °C};$ Table 1; Figure 1	-	-	61.8	A
		$V_{GS} = 10\text{ V}; T_{mb} = 100\text{ °C};$ Table 1	-	-	43.7	A
I_{DM}	peak drain current	$V_{GS} = 10\text{ V}; T_{mb} = 25\text{ °C};$ pulsed; Figure 1	-	-	247	A
P_{tot}	total power dissipation	$T_{mb} = 25\text{ °C}$	-	-	105	W
T_{stg}	storage temperature		-55	-	+175	°C
T_j	junction temperature		-55	-	+175	°C

Source-drain diode

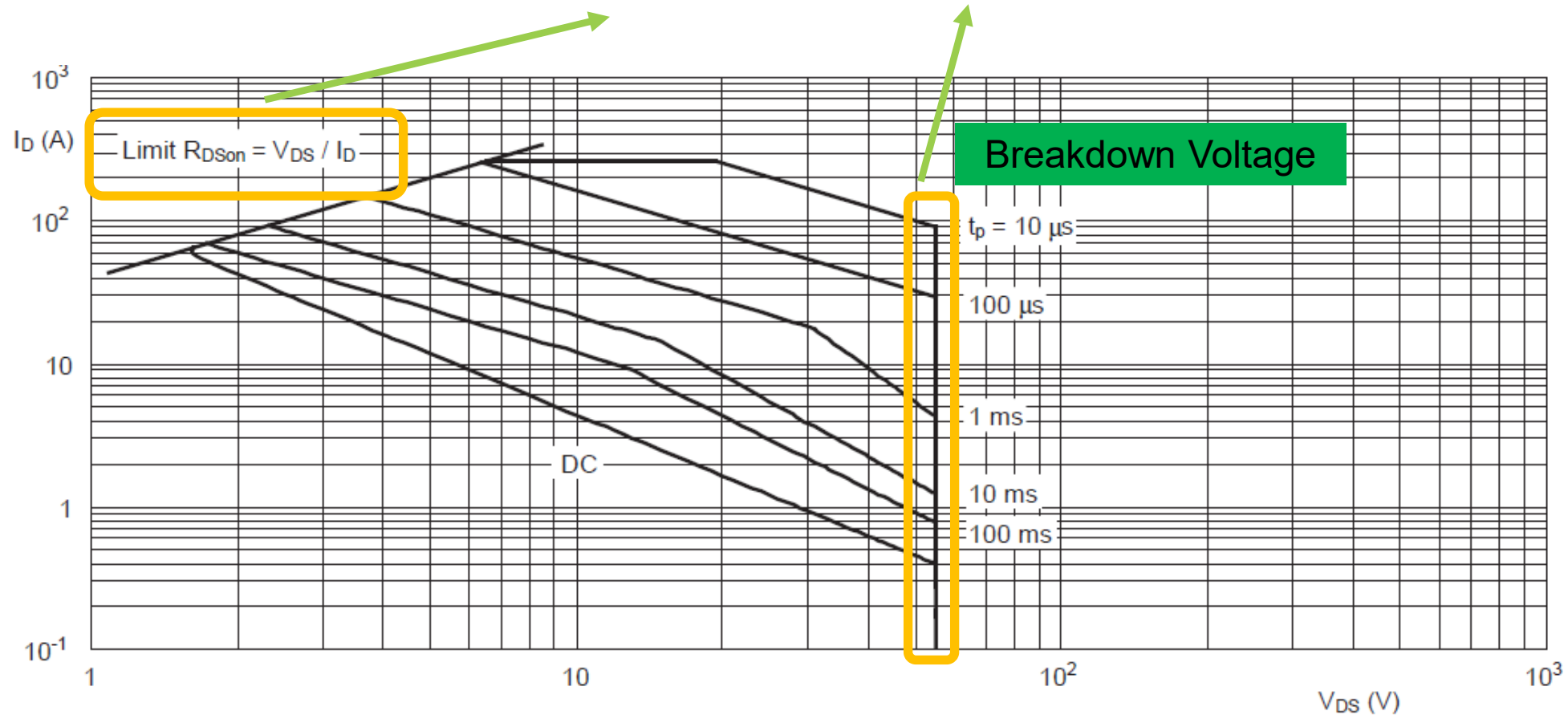
I_S	source current	$T_{mb} = 25\text{ °C}$	-	-	61.8	A
I_{SM}	peak source current	$t_p \leq 10\text{ }\mu\text{s};$ pulsed; $T_{mb} = 25\text{ °C}$	-	-	247	A



Continuous drain current as a function of mounting base temperature

UNDERSTANDING THE DATASHEET OF TRANSISTORS

Two limiting factors: maximum drain current and maximum drain-source voltage



Safe operating area; continuous and peak drain currents as a function of drain-source voltage

Limit R_{DSon} $\longrightarrow$ $P_D = R_{DSon} \cdot I_D^2$

UNDERSTANDING THE DATASHEET OF TRANSISTORS

Table 5. Static characteristics

List of constants and limitations relating to the table i.e. voltages, currents and temperatures

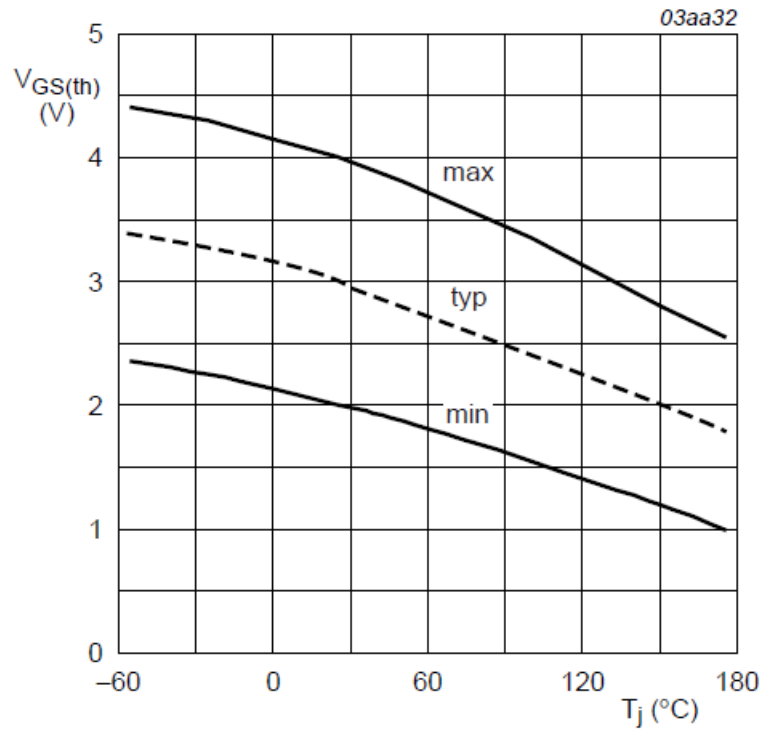
Symbol	Parameter	Conditions	Min	Typ	Max	Unit
$V_{(BR)DSS}$	drain-source breakdown voltage	$I_D = 250 \mu A; V_{GS} = 0 V; T_j = 25 ^\circ C$	55	-	-	V
		$I_D = 250 \mu A; V_{GS} = 0 V; T_j = -55 ^\circ C$	50	-	-	V
$V_{GS(th)}$	gate-source threshold voltage	$I_D = 1 mA; V_{DS} = V_{GS}; T_j = 25 ^\circ C$	2	3	4	V
		$I_D = 250 mA; V_{DS} = V_{GS}; T_j = -55 ^\circ C$	-	-	4.4	V
		$I_D = 1 mA; V_{DS} = V_{GS}; T_j = 175 ^\circ C$	1	-	-	V

Table 5. Static characteristics ...continued

List of constants and limitations relating to the table i.e. voltages, currents and temperatures

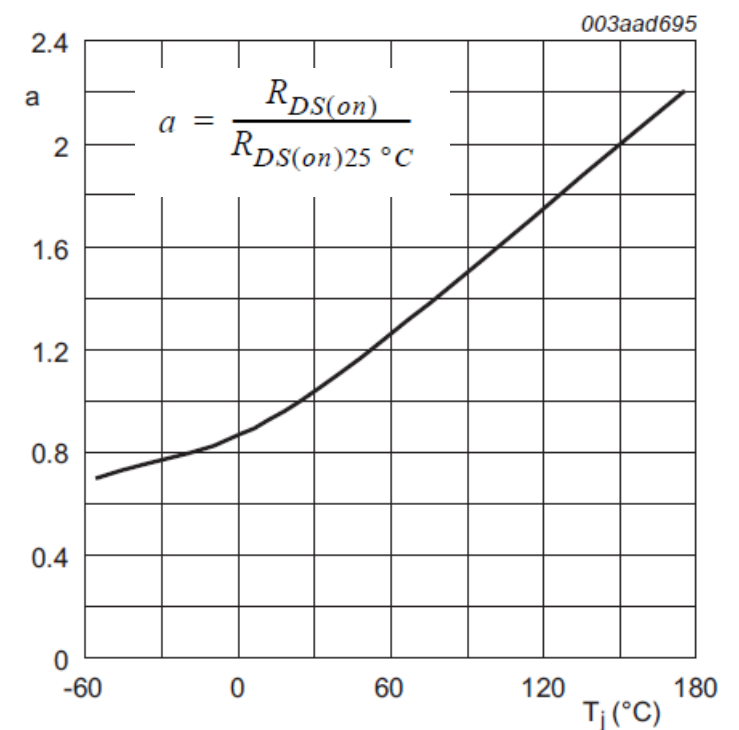
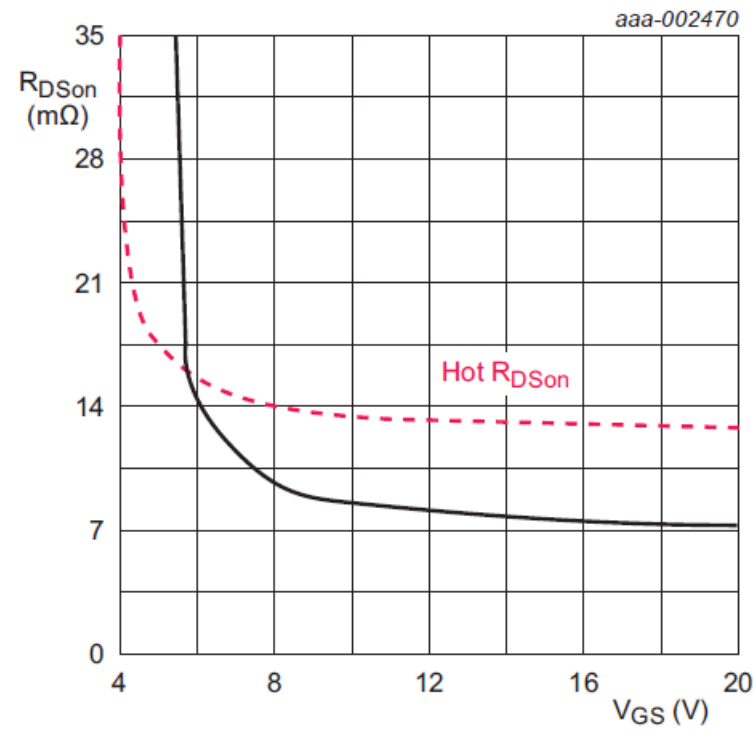
Symbol	Parameter	Conditions	Min	Typ	Max	Unit
I_{DSS}	drain leakage current	$V_{DS} = 55 V; V_{GS} = 0 V; T_j = 25 ^\circ C$	-	0.02	1	μA
		$V_{DS} = 55 V; V_{GS} = 0 V; T_j = 175 ^\circ C$	-	-	500	μA
I_{GSS}	gate leakage current	$V_{DS} = 0 V; V_{GS} = 20 V; T_j = 25 ^\circ C$	-	2	100	nA
		$V_{DS} = 0 V; V_{GS} = -20 V; T_j = 25 ^\circ C$	-	2	100	nA
$R_{DS(on)}$	drain-source on-state resistance	$V_{GS} = 10 V; I_D = 20 A; T_j = 175 ^\circ C;$	-	-	27.6	m Ω
		$V_{GS} = 10 V; I_D = 20 A; T_j = 25 ^\circ C;$	-	8.2	12	m Ω

UNDERSTANDING THE DATASHEET OF TRANSISTORS

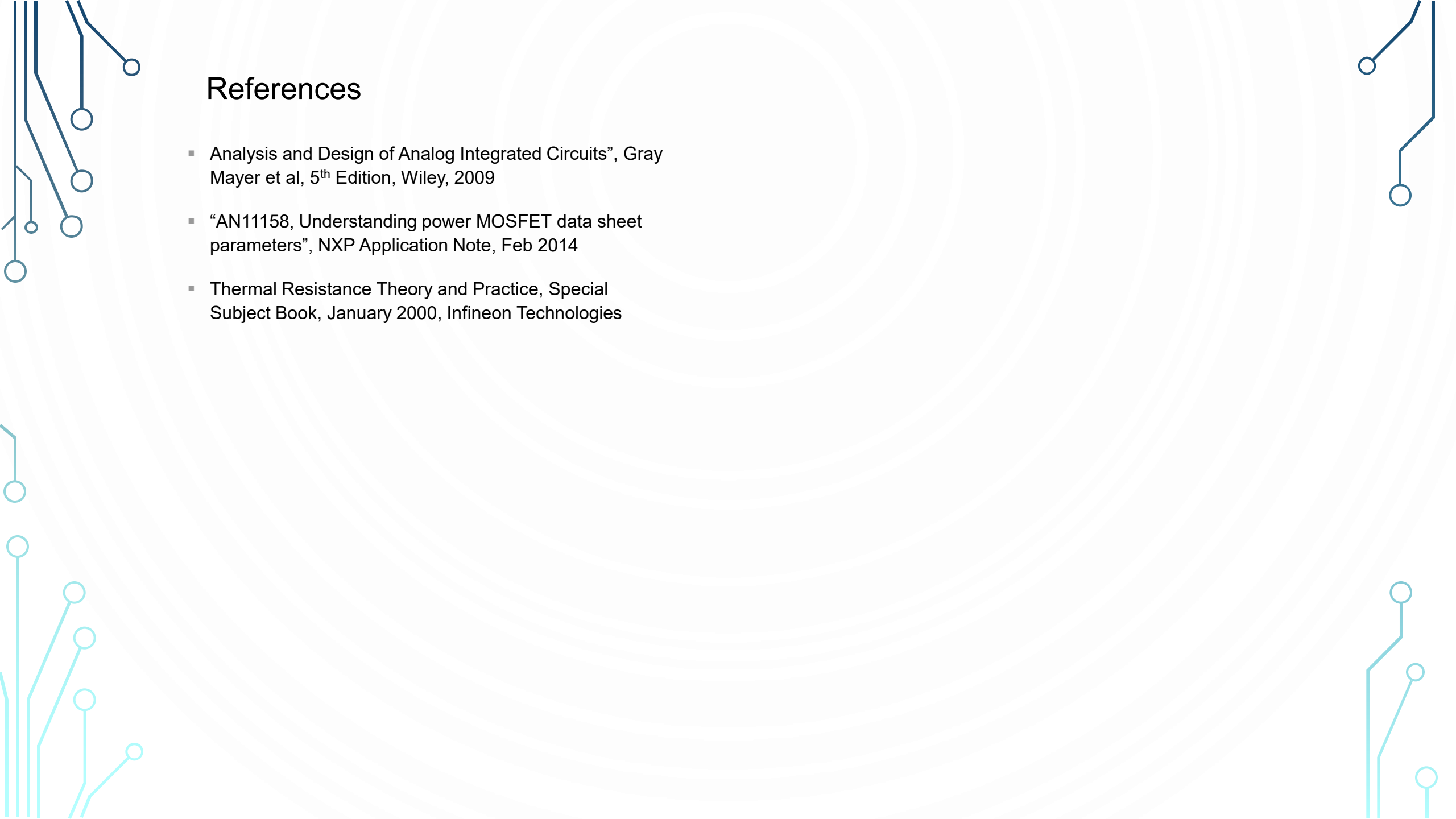


Gate-source threshold voltage as a function of junction temperature

$I_D = 1 \text{ mA}$; $V_{DS} = V_{GS}$



Normalized drain-source on-state resistance factor as a function of junction temperature

The background of the slide features decorative circuit traces in dark blue and light blue. These traces are composed of straight lines and circles, resembling a printed circuit board layout, and are positioned along the top, bottom, and side edges of the slide.

References

- "Analysis and Design of Analog Integrated Circuits", Gray Mayer et al, 5th Edition, Wiley, 2009
- "AN11158, Understanding power MOSFET data sheet parameters", NXP Application Note, Feb 2014
- Thermal Resistance Theory and Practice, Special Subject Book, January 2000, Infineon Technologies